

Complex Integration (5M)

$$\int_0^2$$

$$z_2 + iy_2$$

$$I = \int \underbrace{f(z)}_{\substack{\text{Real} \\ z_1 + iy_1 \\ x \ y}} \underbrace{dz}_{\substack{x \ y \\ x \ y}}$$

Problem: 2 variables & single integration. i.e. dz

Solution: Convert 2 variables into single variable v .
Integration & Interpret

Steps

- ✓ I. Equation →
- ✓ II. Derivative
- III. $f(z)$

Two pt formula

$$A(x_1, y_1) \ B(x_2, y_2)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Ex. 1 Evaluate $\int_0^{1+i} (x-y+ix^2) dz$ along the straight line

$$z = x+iy$$

from $z=0$ to $z=1+i$.

$$\text{iii) } f(z) = x-y+ix^2$$

$$= i(1+i) \int x^2 dx$$

$$= x-x+ix^2 = (i+i^2) \left[\frac{x^3}{3} \right]_0^1$$

$$= (i-1) \left(\frac{1}{3} - 0 \right)$$

$$\boxed{f(z) = ix^2}$$

$$\text{iv) } dz = dx + i dy$$

$$dz = dx + i dx$$

$$\boxed{1 = \frac{i}{3} - \frac{1}{3}}$$

$$\boxed{dz = (1+i) dx}$$

$$\text{v) } I = \int_0^1 ix^2 (1+i) dx$$

$$\frac{y-0}{x-0} = \frac{1-0}{1-0}$$

$$\frac{y}{x} = \frac{1}{1}$$

$$\text{ii) } y = x$$

$$\frac{dy}{dx} = \frac{dx}{dx}$$

$$\frac{dy}{dx} = 1$$

$$\boxed{dy = dx}$$

$$z = 0 \text{ to } z = 1+i$$

$$x \ y \quad x \ y$$

$$(0, 0) \quad (1, 1)$$

$$x_1, y_1 \quad x_2, y_2$$

$$\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$$

$$\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$$

$$\text{Sol: } \rightarrow I = \int_0^{1+i} (x-y+ix^2) dz$$

1) Eqn

ii) dz

iii) f(z)

iv) dz

v) Integr

Ex. 2 Evaluate $\int_0^{3+i} z^2 dz$ along parabola $x=y^2$.

Sol: $I = \int_{0+0i}^{3+i} z^2 dz$

$f(z) = z^2$

$= (x+iy)^2$

$= x^2 + 2xyi + (iy)^2$

$f(z) = x^2 + 2xyi - y^2$

i) $x = y^2$

ii) $\frac{dx}{dy} = \frac{dy^2}{dy}$

$\frac{dx}{dy} = 2y$

$dx = 2y dy$

iii) $f(z) = x^2 + 2xyi - y^2$

$= (y^2)^2 + 2y^2yi - y^2$

$z = x+iy$

$f(z) = y^4 + 2y^3i - y^2$

iv) $dz = dx + i dy$

$= 2y dy + i dy$

$dz = (2y+i) dy$

v) $I = \int_0^1 (y^4 + 2y^3i - y^2)(2y+i) dy$

$= \int_0^1 [2y^5 + 4y^4i - 2y^3 + y^3i - 2y^2 - y^2i] dy$

$I = \int_0^1 [2y^5 + 5y^4i - 4y^3 - y^2i] dy$

$= \left[\frac{2y^6}{6} + \frac{5iy^5}{5} - \frac{4y^4}{4} - \frac{y^3}{3} \right]_0^1$

$= \left[\frac{y^6}{3} + iy^5 - y^4 - \frac{y^3}{3} \right]_0^1$

$= \frac{1}{3} + i - 1 - \frac{i}{3}$

$= (\frac{1}{3} - 1) + (i - \frac{i}{3})$

1) Eqn

ii) dvt

iii) f(z)

iv) dz

v) Integr

Ex 3 Evaluate $\int_0^{1+i} (z^2 - iy) dz$ along straight line

from $z=0$ to $z=1+i$

$$f(z) = z^2 - iz$$

$$= (1+i) \left(\frac{z^3}{3} - i \frac{z^2}{2} \right) \Big|_0^{1+i}$$

$$= (1+i) \left[\frac{1}{3} - \frac{i}{2} \right] - 0$$

$$= \frac{1}{3} - \frac{i}{2} + \frac{i}{3} - \frac{i^2}{2}$$

$$= \left(\frac{1}{3} + \frac{1}{2} \right) + \left(-\frac{i}{2} + \frac{i}{3} \right)$$

$$\boxed{I = \frac{5}{6} - \frac{i}{6}}$$

Sol: $I = \int_0^{1+i} (z^2 - iy) dz$

$$\therefore \frac{y-0}{x-0} = \frac{1-0}{1-0} \quad \text{iv) } dz = dx + i dy$$

$$\frac{y}{x} = \frac{1}{1}$$

$$= dx + i dx$$

$$\boxed{dz = (1+i) dx}$$

i) $y = x$

v) $I = \int_0^1 (x^2 - ix)(1+i) dx$

$$= (1+i) \int_0^1 (x^2 - ix) dx$$

1) Eq

ii) dvt

iii) $f(z)$

iv) dz

v) Integr

$$f(z) = z^2 - iy$$

$z=0$ to $z=1+i$

$x=0, y=0, \quad x=1, y=1$

$(0,0)$

$(1,1)$

x_1, y_1

x_2, y_2

$$\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$$

ii) $\frac{dy}{dx} = \frac{dx}{dx}$

$$dy = dx$$

iii) $f(z) = z^2 - iy$

Examples on Circle

i) ~~$z = x + iy = re^{i\theta}$~~

ii) $|z| = r$

unit circle: $r = 1$

iv) $e^{j\theta} = \cos\theta + j\sin\theta$

$e^{-j\theta} = \cos\theta - j\sin\theta$

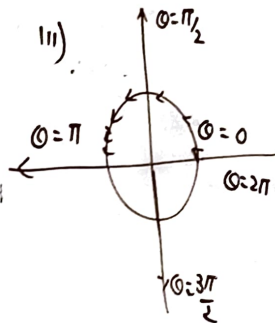
i) $E q^n$

ii) dr/dt

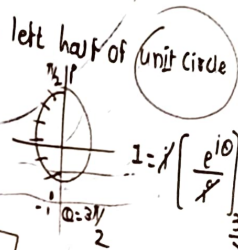
iii) $f(z)$

iv) dz

v) Integr



Ex 4. Evaluate $\int_C |z| dz$, where C is left half of unit circle
from $z = -1$ to $z = i$



sol $\Rightarrow I = \int_C |z| dz$

$\frac{dz}{d\theta} = e^{i\theta} i$

$1 = i \left[\frac{e^{i\theta}}{i} \right]_{\pi}^{\pi/2}$

$= \left[\frac{i\pi}{2} - \frac{i3\pi}{2} \right]$

$= \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) - \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right)$

$= (0 + i) - (0 - i)$

$= 1 + i$

$I = 2i$

$dz = e^{i\theta} i d\theta$

iii) $f(z) = |z| = 1$

iv) $I = \int_{\pi}^{\pi/2} 1 \cdot e^{i\theta} i d\theta$

$= i \int_{\pi}^{\pi/2} e^{i\theta} d\theta$

i) $z = r e^{i\theta}$

unit circle

$|z| = r = 1$

$z = e^{i\theta}$

ii) $\frac{dz}{d\theta} = \frac{d e^{i\theta}}{d\theta}$

~~Eq~~

~~drt~~

~~f(z)~~

dz

v) Integr

Eq

Ex 5. Show that $\int_C \log z \, dz = 2\pi i$, where C is unit circle.

Sol $\rightarrow I = \int_C \log z \, dz$

$f(z) = \log z$

i) $z = re^{i\theta}$

unit circle

$r = 1$

$\therefore z = e^{i\theta}$

ii) $\frac{dz}{d\theta} = \frac{d}{d\theta} e^{i\theta}$

$\frac{dz}{d\theta} = e^{i\theta} i$

$dz = e^{i\theta} i d\theta$

iii) $f(z) = \log z$

$\log e^{i\theta}$

$= i\theta$

$f(z) = i\theta$

$= i\theta$

$I = - \left[\left(\frac{2\pi e^{i2\pi}}{i} + e^{i2\pi} \right) - \left(0 + e^0 \right) \right]$

$= - \left[\frac{2\pi (1)}{i} + 1 - 1 \right]$

$= - \int_0^{2\pi} 0 \cdot e^{i\theta} d\theta$

$= - \left[\left(\frac{0 \times e^{i0}}{i} \right) - \left(\frac{-1 \times (e^{i0})}{i \times 1^0} \right) \right]_0^{2\pi}$

$= - \left[\frac{0 \cdot e^{i0}}{i} + e^{i0} \right]_{0}^{2\pi}$

$= - \frac{2\pi}{i} \times \frac{1}{i}$

$= \frac{-2\pi i}{-1}$

$I = 2\pi i$

~~Eq~~

~~dvt~~

~~f(z)~~

in dz

v) Integr

Taylor's & Laurents Series.

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$$

Ex 1 Expand $f(z) = \frac{z+2}{(z-3)(z-4)}$ about $z=0$.

Indicating ROC in each case.

$\therefore$ (1) becomes

$$f(z) = \frac{-5}{z-3} + \frac{6}{z-4}$$

(II) Conditions

$$z=3, z=4$$

$$|z|=3, |z|=4$$

$$|z| < 3$$

$$3 < |z| < 4$$

$$|z| > 4$$

(III) Expansion: $\frac{|z|}{1} \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array}$ (a) $3 < |z| < 4$

Case (i) $|z| < 3$, $|z| < 4$

$$f(z) = \frac{-5}{z-3} + \frac{6}{z-4}$$

$$= \frac{-5}{3\left(\frac{z}{3}-1\right)} + \frac{6}{4\left(\frac{z}{4}-1\right)}$$

$$= \frac{-5}{-3\left(1-\frac{z}{3}\right)} - \frac{3}{2} \frac{1}{\left(1-\frac{z}{4}\right)}$$

$$= \frac{5}{3} \left(1-\frac{z}{3}\right)^{-1} - \frac{3}{2} \left(1-\frac{z}{4}\right)^{-1}$$

$$= \frac{5}{3} \left[1 + \frac{z}{3} + \frac{z^2}{3^2} + \dots \right] - \frac{3}{2} \left[1 + \frac{z}{4} + \frac{z^2}{4^2} + \dots \right]$$

$$f(z) = \frac{-5}{z-3} + \frac{6}{z-4}$$

$$= \frac{-5}{z\left(1-\frac{3}{z}\right)} + \frac{6}{4\left(\frac{z}{4}-1\right)}$$

$$= \frac{-5}{z\left(1-\frac{3}{z}\right)} - \frac{3}{2} \frac{1}{\left(1-\frac{z}{4}\right)}$$

$$= \frac{-5}{z\left(1-\frac{3}{z}\right)} - \frac{3}{2} \left(1-\frac{z}{4}\right)^{-1}$$

$$= \frac{-5}{z} \left[1 + \frac{3}{z} + \frac{3^2}{z^2} + \dots \right] - \frac{3}{2} \left[1 + \frac{z}{4} + \frac{z^2}{4^2} + \dots \right]$$

Q.1 a) Partial fraction.

$$\frac{z+2}{(z-3)(z-4)} = \frac{A}{z-3} + \frac{B}{z-4} \quad \text{--- (1)}$$

$$z-3=0$$

$$z=3$$

$$A = \left[\frac{z+2}{z-4} \right]_{z=3}$$

$$= \frac{5}{3-4} = -5$$

$$B = \left[\frac{z+2}{z-3} \right]_{z=4}$$

$$= \frac{6}{1} = 6$$

Ex 2 Expand $f(z) = \frac{2z-3}{z^2-4z+3}$ in powers of

$(z-4)$ indicating ROC in each case.

Q21

$$f(z) = \frac{2z-3}{z^2-4z+3}$$

$$B = \left[\frac{2z-3}{z-1} \right]_{z=3} = \frac{3}{2}$$

$$f(z) = \frac{2z-3}{(z-1)(z-3)}$$

a) Partial fraction

$$\frac{2z-3}{(z-1)(z-3)} = \frac{A}{z-1} + \frac{B}{z-3}$$

$$A = \left[\frac{2z-3}{z-3} \right]_{z=1} = \frac{2-3}{1-3} = \frac{1}{2}$$

$$\frac{2z-3}{(z-1)(z-3)} = \frac{1/2}{z-1} + \frac{3/2}{z-3}$$

$$f(z) = \frac{1}{2} \frac{1}{z-1} + \frac{3}{2} \frac{1}{z-3}$$

$$= \frac{1}{2} \frac{1}{(z-4)+1} + \frac{3}{2} \frac{1}{(z-4)+3}$$

$$f(z) = \frac{1}{2} \frac{1}{(z-4)+1} + \frac{3}{2} \frac{1}{(z-4)+1}$$

put $z-4 = x$

$$f(x) = \frac{1}{2} \frac{1}{x+1} + \frac{3}{2} \frac{1}{x+1}$$

Conditions

$$x = -3$$

$$|x| < 3$$

$$x = -1$$

$$|x| < 1$$

$$|x| < 1$$

$$1 < |x| < 3$$

$$|x| > 3$$

iii Expansion

Case (i) $|x| < 1$, $|x| < 3$

$$f(x) = \frac{1}{2} \frac{1}{x+1} + \frac{3}{2} \frac{1}{x+1}$$

$$= \frac{1}{2} \frac{1}{1+(x+1)} + \frac{3}{2} \frac{1}{1+(x+1)}$$

$$= \frac{1}{6} \frac{1}{(1+\frac{x}{3})} + \frac{3}{2} \frac{1}{(1+x)}$$

$$= \frac{1}{6} \left[1 - \frac{x}{3} + \frac{x^2}{3^2} - \dots \right] + \frac{3}{2} \left[1 - x + x^2 - \dots \right]$$

$$f(x) = \frac{1}{6} \left[1 - \frac{x}{3} + \frac{x^2}{3^2} - \dots \right] + \frac{3}{2} \left[1 - x + x^2 - \dots \right]$$

$$f(z) = \frac{1}{6} \left[1 - \frac{(z-4)}{3} + \frac{(z-4)^2}{3^2} - \dots \right] + \frac{3}{2} \left[1 - (z-4) + (z-4)^2 - \dots \right]$$

Q.1

Case-① $1 < |x| < 3$

$1 < |x|, |x| < 3$

$$f(x) = \frac{1}{2} \frac{1}{x+3} + \frac{3}{2} \frac{1}{x+1}$$

$$(1+x)^{-1} = \dots = \frac{1}{2} \frac{1}{3(\frac{x}{3}+1)} + \frac{3}{2} \frac{1}{x(1+\frac{1}{x})}$$

$$= \frac{1}{6} \frac{1}{(1+\frac{x}{3})} + \frac{3}{2x} \frac{1}{(1+\frac{1}{x})}$$

$$= \frac{1}{6} (1+\frac{x}{3})^{-1} + \frac{3}{2x} (1+\frac{1}{x})^{-1}$$

$$f(x) = \frac{1}{6} (1 - \frac{x}{3} + \frac{x^2}{9} - \dots) + \frac{3}{2x} (1 - \frac{1}{x} + \frac{1}{x^2} - \dots)$$

put $z-4=x$

$f(z) = \dots$

Case-② $|x| > 3, |x| > 1$

$$f(x) = \frac{1}{2} \frac{1}{x+3} + \frac{3}{2} \frac{1}{x+1}$$

$$= \frac{1}{2} \frac{1}{x(1+\frac{3}{x})} + \frac{3}{2} \frac{1}{x(1+\frac{1}{x})}$$

$$= \frac{1}{2x} (1+\frac{3}{x})^{-1} + \frac{3}{2x} (1+\frac{1}{x})^{-1}$$

$$f(x) = \frac{1}{2x} \left[1 - \frac{3}{x} + \frac{3^2}{x^2} - \dots \right] + \frac{3}{2x} \left[1 - \frac{1}{x} + \frac{1}{x^2} - \dots \right]$$

$$f(z) = \frac{1}{2(z-4)} \left[1 - \frac{3}{z-4} + \frac{3^2}{(z-4)^2} - \dots \right] + \frac{3}{2(z-4)} \left[1 - \frac{1}{z-4} + \frac{1}{(z-4)^2} - \dots \right]$$

① $|x| < 1$



ii) $1 < |x| < 3$



iii) $|x| > 3$



Q3. Expand $f(z) = \frac{7z-2}{z(z+1)(z-2)}$ about $z = -1$
 Indicating ROC in each case. $(z+1)$

$$f(x) = \frac{1}{x-1} - \frac{3}{x} + \frac{2}{x-3}$$

(case 1) $0 < |x| < 1$
 $0 < |x|, |x| < 1, |x| < 3$

Sol: ① Partial fraction

$$\frac{7z-2}{z(z+1)(z-2)} = \frac{A}{z} + \frac{B}{z+1} + \frac{C}{z-2}$$

$$A = \left[\frac{7z-2}{(z+1)(z-2)} \right]_{z=0} = \frac{-2}{(1)(-2)} = 1$$

$$B = \left[\frac{7z-2}{z(z-2)} \right]_{z=-1} = \frac{-7-2}{(-1)(-3)} = -3$$

$$C = \left[\frac{7z-2}{z(z+1)} \right]_{z=2} = \frac{14-2}{(2)(3)} = 2$$

$$\frac{7z-2}{z(z+1)(z-2)} = \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2}$$

$$f(z) = \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2}$$

$$= \frac{1}{(z+1)-1} - \frac{3}{z+1} + \frac{2}{(z+1)-3}$$

$$f(z) = \frac{1}{(z+1)-1} - \frac{3}{z+1} + \frac{2}{(z+1)-3}$$

put $z+1 = x$

② Conditions

$$x=1, x=0, x=3$$

$$|x|=1, |x|=0, |x|=3$$

$$0, 1, 3$$

$$|x| < 0 \text{ (wrong)}$$

$$i) 0 < |x| < 1$$

$$ii) 1 < |x| < 3$$

$$iii) |x| > 3$$

$$f(x) = \frac{1}{x-1} - \frac{3}{x} + \frac{2}{x-3}$$

$$= \frac{1}{1(x-1)} - \frac{3}{x} + \frac{2}{3(\frac{x}{3}-1)}$$

$$= \frac{1}{-(1-x)} - \frac{3}{x} - \frac{2}{3(1-\frac{x}{3})}$$

$$= -\frac{1}{(1-x)} - \frac{3}{x} - \frac{2}{3(1-\frac{x}{3})}$$

$$f(x) = -\left[\frac{1}{1-x} + \frac{3}{x} + \frac{2}{3\left(1-\frac{x}{3}\right)} \right]$$

$$\text{put } x = z+1$$

$$f(z) =$$

Case II
 $|x| > 0$, $1 < |x|$, $|x| < 3$

$$f(x) = \frac{1}{x-1} - \frac{3}{x} + \frac{2}{x-3}$$

$$= \frac{1}{x(1-\frac{1}{x})} - \frac{3}{x} + \frac{2}{3(\frac{x}{3}-1)}$$

$$= \frac{1}{x(1-\frac{1}{x})} - \frac{3}{x} - \frac{2}{3} \frac{1}{(1-\frac{x}{3})}$$

$$= \frac{1}{x} (1-\frac{1}{x})^{-1} - \frac{3}{x} - \frac{2}{3} (1-\frac{x}{3})^{-1}$$

$$f(x) = \frac{1}{x} \left[1 + \frac{1}{x} + \frac{1}{x^2} \dots \right] - \frac{3}{x} - \frac{2}{3} \left[1 + \frac{x}{3} + \frac{x^2}{3^2} \dots \right]$$

put $x = z+1$

Case III
 $|x| > 3$, $|x| > 1$, $|x| > 0$

$$f(x) = \frac{1}{x-1} - \frac{3}{x} + \frac{2}{x-3}$$

$$= \frac{1}{x(1-\frac{1}{x})} - \frac{3}{x} + \frac{2}{x(1-\frac{3}{x})}$$

$$= \frac{1}{x} (1-\frac{1}{x})^{-1} - \frac{3}{x} + \frac{2}{x} (1-\frac{3}{x})^{-1}$$

$$f(x) = \frac{1}{x} \left[1 + \frac{1}{x} + \frac{1}{x^2} \dots \right] - \frac{3}{x} + \frac{2}{x} \left[1 + \frac{3}{x} + \frac{3^2}{x^2} \dots \right]$$

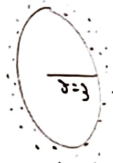
put $x = z+1$

ROC

i) $0 < |x| < 1$



iii) $|x| > 3$



ii) $1 < |x| < 3$



Ex 4. Expand $f(z) = \frac{z^2 - 1}{z^2 + 5z + 6}$ about $z=0$

Indicating ROC in each case

Soln:

$$f(z) = \frac{z^2 - 1}{z^2 + 5z + 6}$$

$\swarrow \begin{matrix} z = -2 \\ z = -3 \end{matrix}$

$$f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$$

DIVISOR

$$\begin{array}{r|l} 1 & \cancel{z^2} - 1 \\ z^2 + 5z + 6 & \cancel{z^2} + 5z + 6 \\ \hline & -5z - 7 \rightarrow R \end{array}$$

$$f(z) = Q + \frac{P}{D}$$

$$= 1 + \frac{-5z - 7}{z^2 + 5z + 6}$$

$$= 1 + \frac{-5z - 7}{(z+2)(z+3)}$$

$$\frac{-5z - 7}{(z+2)(z+3)} = \frac{A}{z+2} + \frac{B}{z+3}$$

$$A = \left. \frac{-5z - 7}{z+3} \right|_{z=-2} = \frac{10-7}{1} = 3$$

$$B = \left. \frac{-5z - 7}{z+2} \right|_{z=-3} = \frac{15-7}{-1} = -8$$

$$\frac{-5z - 7}{(z+2)(z+3)} = \frac{3}{z+2} - \frac{8}{z+3}$$

$$f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$$

HW